# Crop Identification in Topographically Varied Complex Areas in the Loess Plateau Based on Multifeature Integration of Sentinel-2 and ALOS DEM

Qiaomei Su, Weili Zeng, and Jinlong Fan

***Abstract*—Crop identification with remote sensing images has currently emerged as a prominent research topic with a focus on flat regions dominated by crop cultivation. However, farmland is also distributed in vast complex terrain areas where the study of crop identification is necessary and of importance. This study aims to propose an innovative method to address the challenges by investigating what complex terrain poses to the feature extraction and classification algorithms. Shouyang Loess Hill and Jinzhong Basin were chosen as the study area. Three indices, topographic indices, vegetation indices, and texture indices were extracted from ALOS DEM and time-series Sentinel-2 spectral bands, respectively. Various feature combinations of spectral bands and three indices were ingested into the algorithm of random forest (RF), and then five models were built to investigate the impact of various feature combinations on the accuracy of crop identification in both hilly and flat areas. The results indicate that the combination of multiple features can significantly improve the accuracy of crop identification in complex terrain areas, while the differences in flat regions were less pronounced. This study revealed that the combination of diverse features with topographic features is an effective method to improve the accuracy of crop identification in complex terrain areas. The high-accuracy crop maps can provide valuable insights for optimizing agricultural practices and resource management in complex terrain areas. The future study should extend this method to other, more complex terrain areas with efficient optimization of topographic features.**

***Index Terms*—ALOS DEM, complex terrain area, crop identification, feature selection, multiple features combination, Sentinel-2.**

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## I. Introduction

IN RECENT years, there have been extensive studies on the topics of sustainable agricultural development and food security [1], [2], [3]. The information about crop planting in the given administrative unit is the basis for effectively and efficiently conducting agricultural production activities in China [4]. Traditionally, crop cultivation data are routinely reported from local units, so this approach is labor-costing and time-consuming, and susceptible to a large uncertainty with subjective guesswork [5]. Satellite imagery carries ground information on a large-scale and over a long period. Crops can be extracted objectively and accurately from satellite images, and therefore, crop identification became a key research topic in the field of agricultural remote sensing [6], [7], [8].

Many crop identification studies initially used remote sensing images at medium to low resolution, working on a relatively large scale [9]. Wardlow and Egbert [10] used multiple temporal MODIS data to identify America's large-scale farmland. Brown et al. [11] evaluated the changes in crop cultivation in multiple years in Mato Grosso using MODIS-NDVI data. However, crop types can be identified with time-series MODIS data, but its low-resolution limits the accuracy in regions with small field parcels.

Currently, high-resolution images are gradually becoming major data sources for crop mapping [12], [13], [14]. Yu and Shang [15] mapped the primary crop distributions for several years in China's Hetao irrigation zones using HJ-1A/1B imagery, and Fan et al. [16] achieved satisfactory crop classification accuracy in the Yellow River irrigation area with Sentinel-2 optical satellite images.

High-resolution imagery contains and provides not only spectral information but also the spatial structure features and the texture features of ground objects that enhance the crop information extraction [17], [18]. Fan et al. [19] combined spectral features from Sentinel-2 data with a random forest (RF) classifier to map land cover of the Grand Canyon in the Nujiang river, demonstrating that integrating multiple features significantly improves classification accuracy compared to a single feature.

Incorporating diverse features, such as topographic attributes, spectral data, texture characteristics, and vegetation indices has proven to enhance classification accuracy

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[20]. However, including too many features can lead to dimensionality issues and information redundancy, ultimately reducing the final accuracy [21]. Identifying the number of features and optimizing their combinations is therefore crucial to achieve the highest classification accuracy [22].

The extensive use of high-resolution imagery for crop mapping has happened in plain areas, while few studies have explored its application in complex terrain regions, which constitute two-thirds of China's land area. The rugged topography and unique hydrothermal conditions bring underexplored challenges.

To address this gap, this study proposes a crop identification method for complex terrain areas through the comparison by optimizing feature combinations using Sentinel-2 satellite images and ALOS DEM data. Shouyang Loess Hill and Jinzhong Basin in Shanxi province were selected as the study areas. Three kinds of indices, representing topographic features, texture features, and vegetation index features, were extracted with ALOS DEM and time series Sentinel-2 spectral bands, respectively. The feature optimization was performed to address dimensional challenges, and five RF models were constructed by combining spectral bands with various features. Then, the feature selection and the classification accuracy across five RF models were compared and investigated.

The approach not only combines diverse features but also optimizes feature selection to address dimensionality issues, aiming to improve crop identification accuracy in rugged terrain. The results can provide support for the management of crop cultivation in complex terrain areas in the near future.

## II. Materials and Methods

### A. Study Area

The study area lies in the center of Shanxi Province, the middle of northern China, and the eastern part of the Loess Plateau. Shanxi province has limited plains and a large complex terrain, where the mountains and hills account for 80%. The topography of Jinzhong Basin is rather flat, primarily shaped by sediments of the Fen River, which is the second tributary of the Yellow River. Shouyang Loess Hill is a typical area of the Loess Plateau region and is located in the eastern part of Jinzhong Basin. The climate of the research area is characterized by four distinct seasons in a year, sufficient light conditions, and a large temperature difference between night and day. The annual average temperature across Shanxi province is in a range from 4.2 °C to 14.2 °C, and the annual average precipitation is in a range between 358 and 621 mm. The cultivated land of the study area is highly fragmented. Fig. 1 depicts the two study sites, A and B, standing for parts of Jinzhong Basin and Shouyang Loess Hill, respectively.

### B. Satellite Imagery

The satellite data include Sentinel-2 images and ALOS DEM. Five dates of Sentinel-2 data were downloaded from the Copernicus Data Centre with good quality (shown in Table I) from April to August 2023 in the study area. The Sentinel-2 image has 13 spectral bands, and its swath is 290 km. The spatial resolutions vary at 10, 20, and 60 m. For instance, B01, B09, and B10 are in 60-m spatial resolution, and B02, B03, B04, and B08 are in a finer resolution of 10 m. In addition, the other bands, including B05, B06, B07, B08A, B11, and B12 maintain a resolution of 20 m. The released Sentinel-2 data is geometrically, radiometrically, and atmospherically corrected. Four bands with 10-m resolutions were resampled to 20 m, and then all the 20-m bands were selected for this study. Three bands in 60 m were not used in the study.

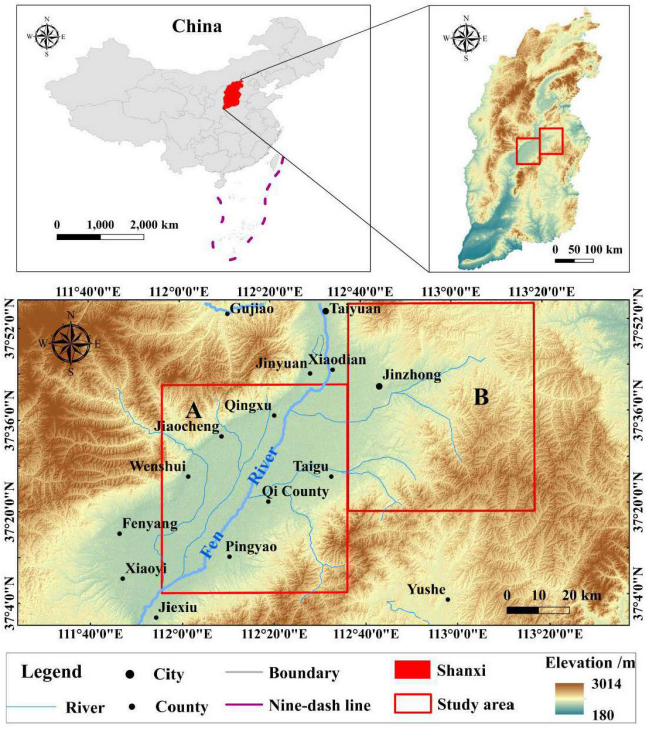



Fig. 1. Sites of the study area.

TABLE I
Used Sentinel-2 Data

| NO | Date | Satellite type | Band number | Cloud and Shade cover /% |
|---|---|---|---|---|
| 1 | 20230430 | S2A | 13 | 1.13 |
| 2 | 20230515 | S2B | 13 | 1.86 |
| 3 | 20230609 | S2A | 13 | 0.51 |
| 4 | 20230704 | S2B | 13 | 0.06 |
| 5 | 20230813 | S2B | 13 | 13.25 |

The ALOS DEM with a spatial resolution of 12.5 m has been obtained from the official website of NASA Earth Observation Data (https://urs.earthdata.nasa.gov). To match the spatial resolution of Sentinel-2 used in this study, the spatial resolution of ALOS DEM was resampled from the original 12.5–20 m, using the nearest neighbor sampling.

### C. Training Sample Data

In the study, the training sample data were mainly collected in the field and further added by visually interpreting based on the valid images and referencing the Google Earth images. On 28–29 May and 17–19 July 2023, field campaigns were

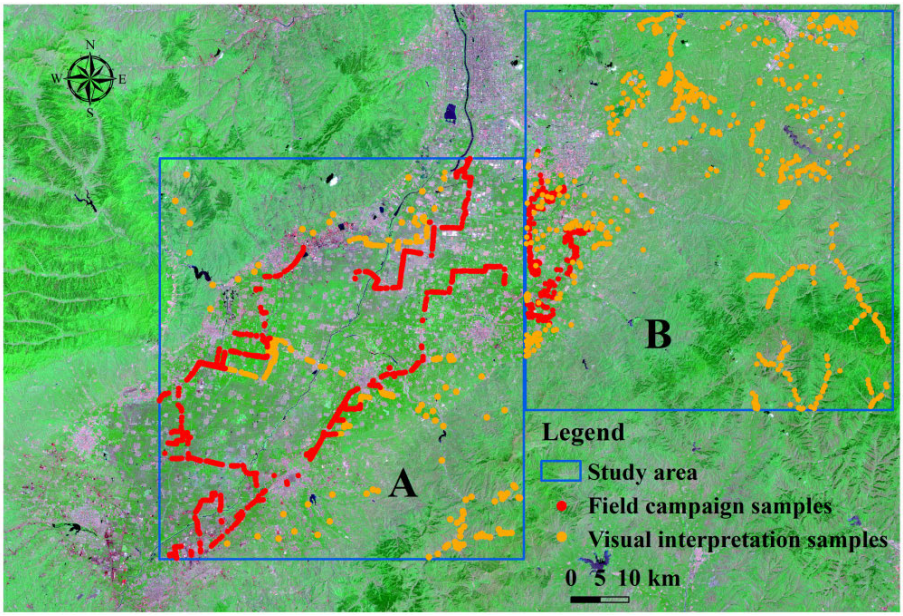



Fig. 2. Distribution of sample sites on Sentinel-2 image.

conducted. In collecting field data, GPS cameras were used to take sample photographs along roads in the study area. The photographs were further interpreted visually with a photograph data interpretation tool, and finally, the crop type and other land cover types with the coordinates of latitude and longitude were output to a formatted text file that contains the corresponding class, class code, collecting time and date, the corresponding photograph filename, and roadside. Supported with field photographs, training sample data for crop identification were further prepared visually. In the process of adding more samples, the profile of potential samples for each crop type was checked with the time series of satellite images and compared with known samples. The similarity principle was then applied to the patches to be selected as the samples. In this process, it also took the evenly spatial distribution of samples into consideration to have relatively spatially distributed and balanced sample sites. Finally, about 3582 sample sites were collected in the field, and 1316 sample sites were further obtained through visual interpretation on the latest images. Finally, in study area A, 2392 samples were collected in the field under an altitude of 1000 m, while 406 samples under the altitude of 1000 m were interpreted, and 74 samples above the altitude of 1000 m were interpreted. In study area B, 558 samples were collected in the field under the altitude of 1000 m, while 205 samples under the altitude of 1000 m were interpreted, 384 samples between the altitude of 1000–1500 m, and 879 samples above the altitude of 1500 m were interpreted. 70% of all samples were randomly separated as samples for model training, while the rest 30% was for the validation. Fig. 2 displays the distribution of sample sites within the study area.

## III. Method

### A. Study Flowchart

The flowchart of the research is shown in Fig. 3: 1) all the features (topographic features, texture features, and vegetation index) were extracted and analyzed; 2) for the sake of exploring how the different feature combinations affect the accuracy of crop identification, five models were proposed; and 3) RF algorithm was selected to identify the crop types, and then the accuracy of different feature combinations for the classification were evaluated and compared between Shouyang Loess Hill and Jinzhong Basin areas.

### B. Feature Extraction

Due to the rapid improvement of the spatial resolution of satellite images, multiple features of the images become more prominent, the colors become fuller, the textures become richer, and the shapes become more prominent [23], [24]. In the study, topographic features, spectral features, vegetation index, and texture features have been used, and the selected image, dated July 4, 2023, in the critical period of crop identification, was used to extract feature variables, and a comprehensive set of 32 feature variables was chosen for multifeature combination analysis.

*1) Spectral Characteristics Extraction:* Different types of crops have different developmental patterns and growth phenological information, so the radiation characteristics and spectral reflectance of different crops vary and finally present different brightness values in optical images. A total of ten spectral features have been selected and processed with Sentinel-2 data, including tricolor (B02, B03, and B04), red edge vegetation Edge1, vegetation Edge2, vegetation Edge3 (B05, B06, and B07), near-infrared band NIR, NIR arrow (B08 and B08A), shortwave infrared band SWIR1, and SWIR2 (B11 and B12).

*2) Vegetation Index Extraction:* Ten types of vegetation index features [25], [26], [27] were selected and extracted according to their calculation formulas. The vegetation index and its calculation formulas are shown in Table II [formula (1)–(10)], where $\rho_$red, $\rho_$green, $\rho_$blue, $\rho_$nir, $\rho_$swir1, and $\rho_$swir2 represent the reflectance of tricolor bands, NIR,

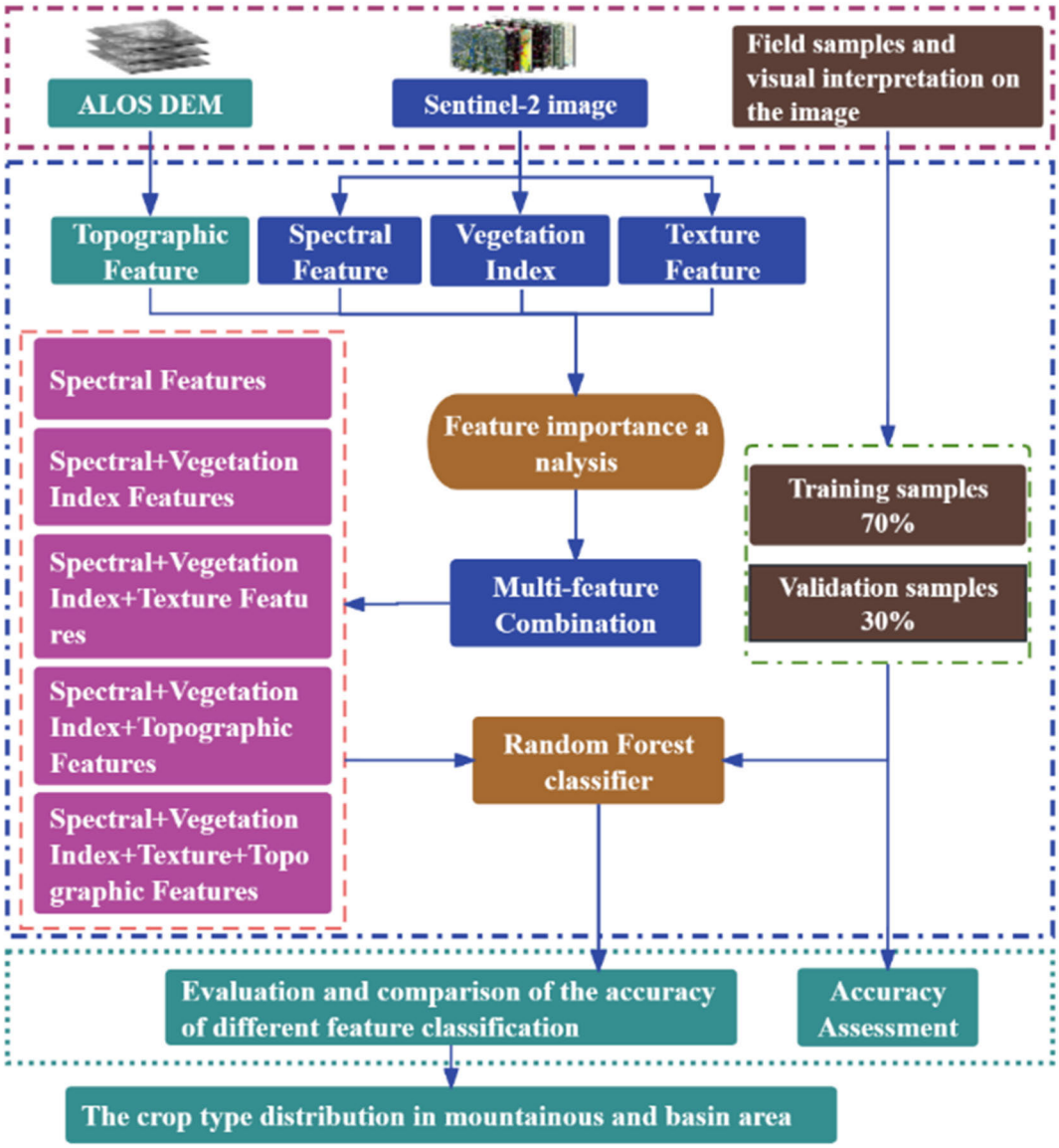



Fig. 3. Flowchart of the study.

SWIR1, and SWIR2 data of the Sentinel-2 images, respectively.

*3) Texture Features Extraction:* Texture information is generated, representing the spatial variation of the image. In this study, the principle of the gray-level co-occurrence matrix (GLCM) [27] has been adopted for calculating texture features of the image. The GLCM is a commonly used algorithm to measure the texture features of the remote sensing data. Through several experiments and comparative analyses, based on the horizontal direction, a $3 \times 3$ pixel moving window, and a step size of one on the near-infrared band (B08), eight texture features were calculated with GLCM. The texture features and their formulas are shown in Table III [see formulas (11)–(18)].

In these formulas, N represents the gray scale level of texture images; $P_{i,j}$ is the element values of normalized GLCM at the position (i, j); i and j stand for the rows and columns in the GLCM, respectively; $Mea_r$ and $Mea_c$ represent the average values of the rows and columns, respectively; $Var_r$ and $Var_c$ are the variances of rows and columns, respectively. Texture feature images are shown in Figs. 4 and 5.

*4) Topographic Feature Extraction:* Topographic features are important indicators for complex terrain areas. The ALOS DEM data were processed in the ARCGIS platform. The four topographic feature information, like elevation, slope angle, slope aspect, and hill shade data of the study area, were extracted. Figs. 6 and 7 represent the topographic characteristics of the two regions, respectively.

### C. Modeling Strategy

Five models were designed to investigate the impact of various feature combinations on the accuracy of crop identification in Shouyang Loess Hill and Jinzhong Basin areas. Model A: traditional spectral features only; model B: spectral

TABLE II
VEGETATION INDEX AND ITS FORMULAS

| Vegetation index | The calculation formulas | Serial Number |
|---|---|---|
| Normalized Difference Vegetation Index (NDVI) | $\boldsymbol{NDVI = \frac{\rho_{nir} - \rho_{red}}{\rho_{nir} + \rho_{red}}}$ | (1) |
| Normalized Difference Water Index (NDWI) | $\boldsymbol{NDWI = \frac{\rho_{green} - \rho_{nir}}{\rho_{green} + \rho_{nir}}}$ | (2) |
| Land Surface Water Index (LSWI) | $\boldsymbol{LSWI = \frac{\rho_{nir} - \rho_{swir1}}{\rho_{nir} + \rho_{swir1}}}$ | (3) |
| Modified Soil Adjusted Vegetation Index (MSAVI) | $\boldsymbol{MSAVI = \frac{2\rho_{nir} + 1 - \sqrt{(2\rho_{nir} + 1)^2 - 8(\rho_{nir} - \rho_{red})}}{2}}$ | (4) |
| Ratio Vegetation Index (RVI) | $\boldsymbol{RVI = \frac{\rho_{red}}{\rho_{nir}}}$ | (5) |
| Difference Vegetation Index (DVI) | $\boldsymbol{DVI = \rho_{nir} - \rho_{red}}$ | (6) |
| Renormalized Difference Vegetation Index (RNDVI) | $\boldsymbol{RNDVI = \frac{NDVI - NDVI_{min}}{NDVI_{max} - NDVI_{min}} \times 255}$ | (7) |
| Atmospheric Resistant Vegetation Index (ARVI) | $\boldsymbol{ARVI = \frac{\rho_{nir} - (2\rho_{red} + \rho_{blue})}{\rho_{nir} + (2\rho_{red} + \rho_{blue})}}$ | (8) |
| Normalized Difference Tillage Index (NDTI) | $\boldsymbol{NDTI = \frac{\rho_{swir1} - \rho_{swir2}}{\rho_{swir1} + \rho_{swir2}}}$ | (9) |
| Enhanced Vegetation Index (EVI) | $\boldsymbol{EVI = \frac{2.5(\rho_{nir} - \rho_{red})}{\rho_{nir} + 6\rho_{red} - 7.5\rho_{blue} + 1}}$ | (10) |

TABLE III
TEXTURE FEATURES AND THEIR FORMULA

| Texture | The calculation formulas | Serial Number |
|---|---|---|
| Mean (Mea) | $\boldsymbol{Mea = \sum_{i,j=0}^{N-1} i \cdot P_{i,j}}$ | (11) |
| Variance (Var) | $\boldsymbol{Var = \sum_{i,j=0}^{N-1} i \cdot P_{i,j}(i - Mea)^2}$ | (12) |
| Homogeneity (Hom) | $\boldsymbol{Hom = \sum_{i,j=0}^{N-1} \frac{i \cdot P_{i,j}}{1+(i-j)^2}}$ | (13) |
| Contrast (Con) | $\boldsymbol{Con = \sum_{i,j=0}^{N-1} (i - j)^2 \cdot P_{i,j}}$ | (14) |
| Dissimilarity (Dis) | $\boldsymbol{Dis = \sum_{i,j=0}^{N-1} i \cdot P_{i,j}|i - j|}$ | (15) |
| Entropy (Ent) | $\boldsymbol{Ent = \sum_{i,j=0}^{N-1} i \cdot P_{i,j} \ln P_{i,j}}$ | (16) |
| Angular Second Moment (ASM) | $\boldsymbol{ASM = \sum_{i,j=0}^{N-1} i \cdot (P_{i,j})^2}$ | (17) |
| Correlation (Cor) | $\boldsymbol{Cor = \sum_{i,j=0}^{N-1} i \cdot P_{i,j} \frac{(i - Mea_r)(j - Mea_c)}{\sqrt{Var_r \cdot Vaar_c}}}$ | (18) |

features and vegetation index; model C: spectral bands, vegetation index, and texture features; model D: spectral bands, vegetation index, and topographic features; model E: spectral, vegetation index, texture, and topographic features. All these different models were set up with the following two main purposes: 1) to assess the significance of various feature factors and investigate the influence of various feature variables on crop information extraction and 2) to determine the most

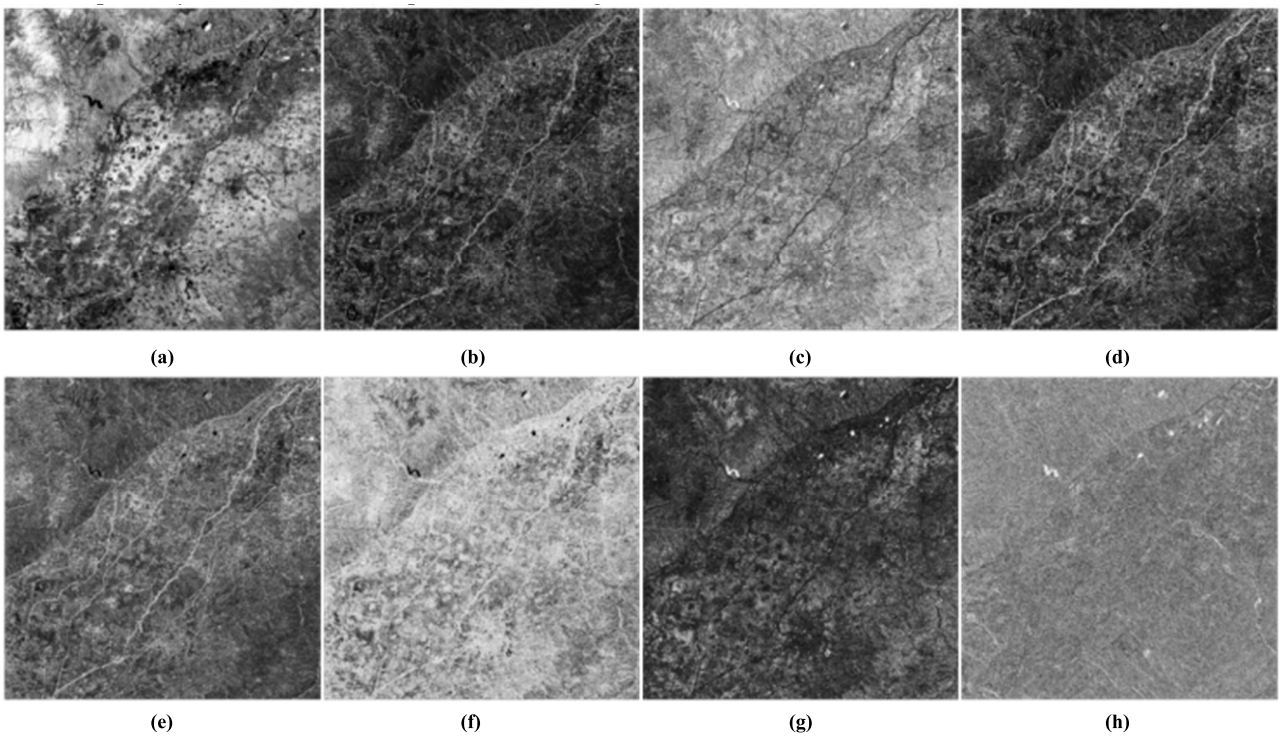



Fig. 4. Texture feature images of Jinzhong Basin. (a) Mean. (b) Variance. (c) Homogeneity. (d) Contrast. (e) Dissimilarity. (f) Entropy. (g) Angular second moment. (h) Correlation.

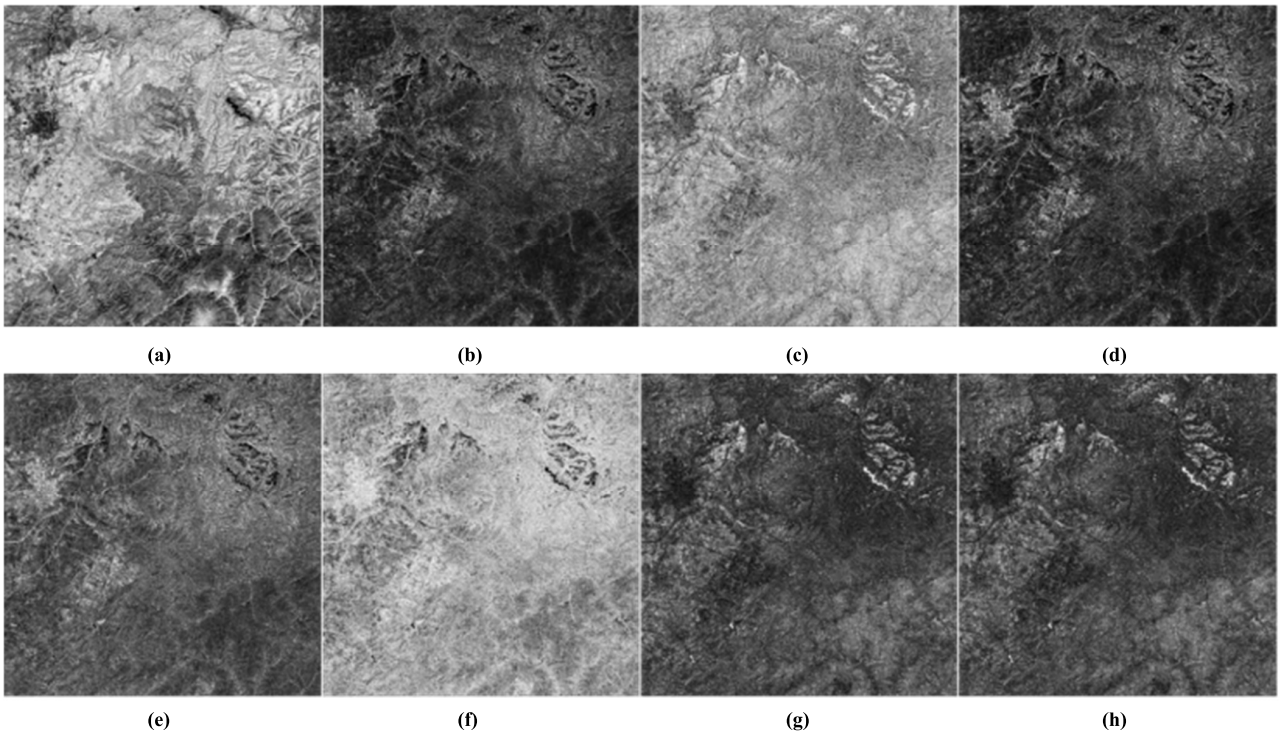
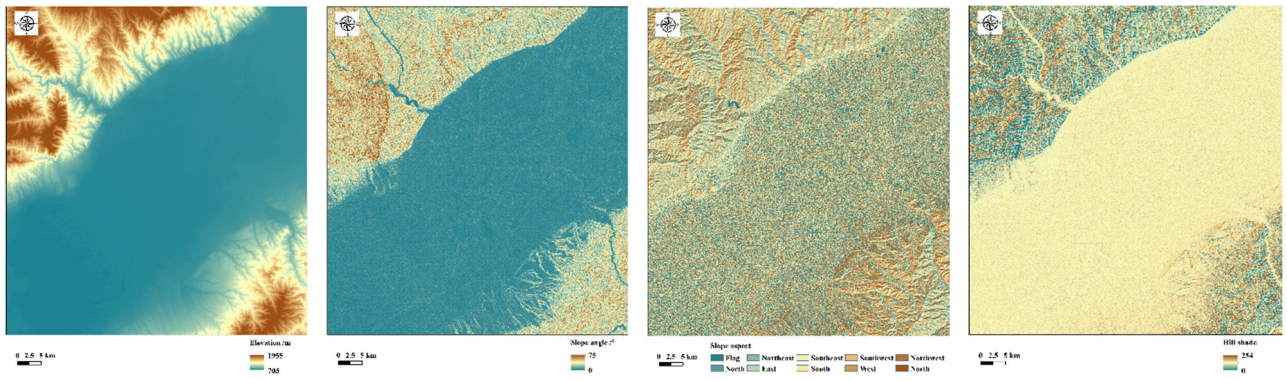



Fig. 5. Texture feature images of Shouyang Loess Hill. (a) Mean. (b) Variance. (c) Homogeneity. (d) Contrast. (e) Dissimilarity. (f) Entropy. (g) Angular second moment. (h) Correlation.

effective means of enhancing crop extraction accuracy via model comparisons.

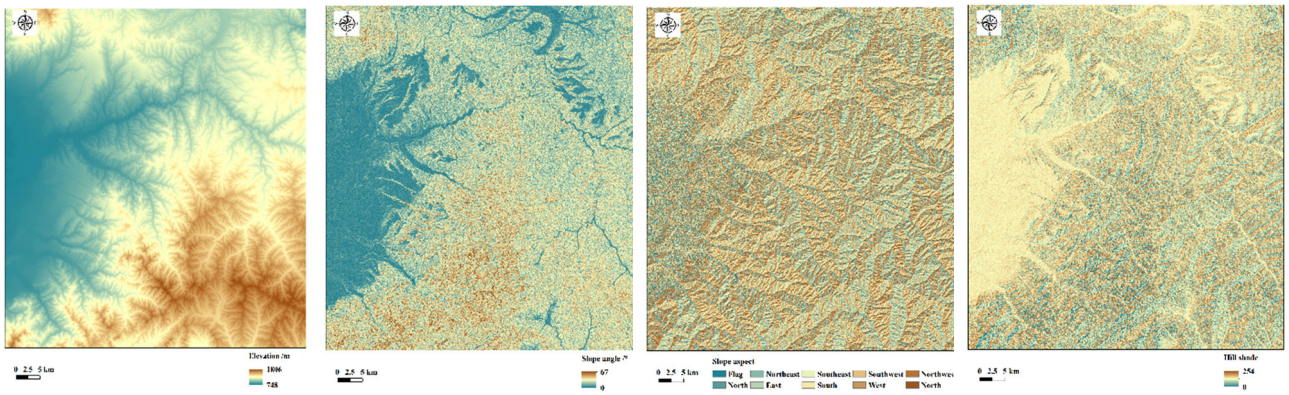
### *D. Feature Optimization Algorithm*

In this study, the RF algorithm has been established by invoking the Scikit-learn library on the Python platform [12]. The value of the significance score of each feature variable was calculated in the RF model using the method of average impurity reduction and sorted in descending order. Usually, the higher the significance score of each feature variable, the greater the contribution of the feature to the classification result of the model. According to the order of significance,



(a) (b) (c) (d)

Fig. 6. Topographic features of Jinzhong Basin. (a) Elevation. (b) Slope angle. (c) Slope aspect. (d) Hill shade.



(a) (b) (c) (d)

Fig. 7. Topographic features of Shouyang Loess Hill. (a) Elevation. (b) Slope angle. (c) Slope aspect. (d) Hill shade.

TABLE IV
ACCURACY INDICATORS AND THEIR FORMULA

| Vegetation index | The calculation formulas | Serial Number |
|---|---|---|
| OA | $\boldsymbol{OA} = \frac{\sum_{i=1}^{k} N_{ii}}{\sum_{i=j=1}^{k} N_{ii}}$ | (19) |
| Kappa | $\boldsymbol{Kappa} = \frac{N\sum_{i=1}^{k} N_{ii} - \sum_{i=1}^{k} (N_{i+}) \cdot (N_{+i})}{N^2 - \sum_{i=1}^{k} (N_{i+}) \cdot (N_{+i})}$ | (20) |
| F1 | $\boldsymbol{F1} = \frac{2N_{ii}}{(N_{i+}) + (N_{+i})}$ | (21) |

the first feature was selected to test first, and then the second feature. Assuming N feature variables in total, N RF models with different combinations can be obtained by means of the sequence, and the out-of-bag (OOB) errors were calculated, respectively. After comprehensive consideration of model accuracy and model complexity, the optimal feature combinations were determined.

### E. Accuracy Measures

The confusion matrix, derived from the validation samples, was computed to assess classification accuracy. The accuracy indicators, such as overall accuracy (OA), kappa coefficient (kappa), and $F1$-score ($F1$), were employed to provide quantitative measures of the accuracy of the classified results [28]. OA denotes the proportion of correctly classified samples to the total samples. Kappa serves as an indicator of classification accuracy, typically ranging from 0 to 1. $F1$ represents the reconciled average cartographic accuracy value and user accuracy, which can be used to generally evaluate the output results of the algorithm. The value range of the $F1$ score is around 0 ~ 1, and the closer the $F1$ is to 1, the better the model will be. In this study, kappa and $F1$ values were converted to percentages to keep in line with OA. The accuracy indicators and their formula are shown in Table IV [see formulas (19)–(21)].

In the formulas, generally N stands for the total number of samples; k means the number of class types of the confusion matrix; Nii represents the number of samples classified correctly and represented by column i corresponding to row i in the confusion matrix; Ni+ and N+i represent the sum of all samples in the *i*th row and *i*th column of the confusion matrix, respectively.

## IV. RESULTS

### A. Classification Results

The RF model, combined with multiple feature variables, was used to construct different feature combination schemes and then to extract crop types in complex terrain areas and flat areas, respectively. The results of crop type extraction with different feature combinations are shown in Figs. 8

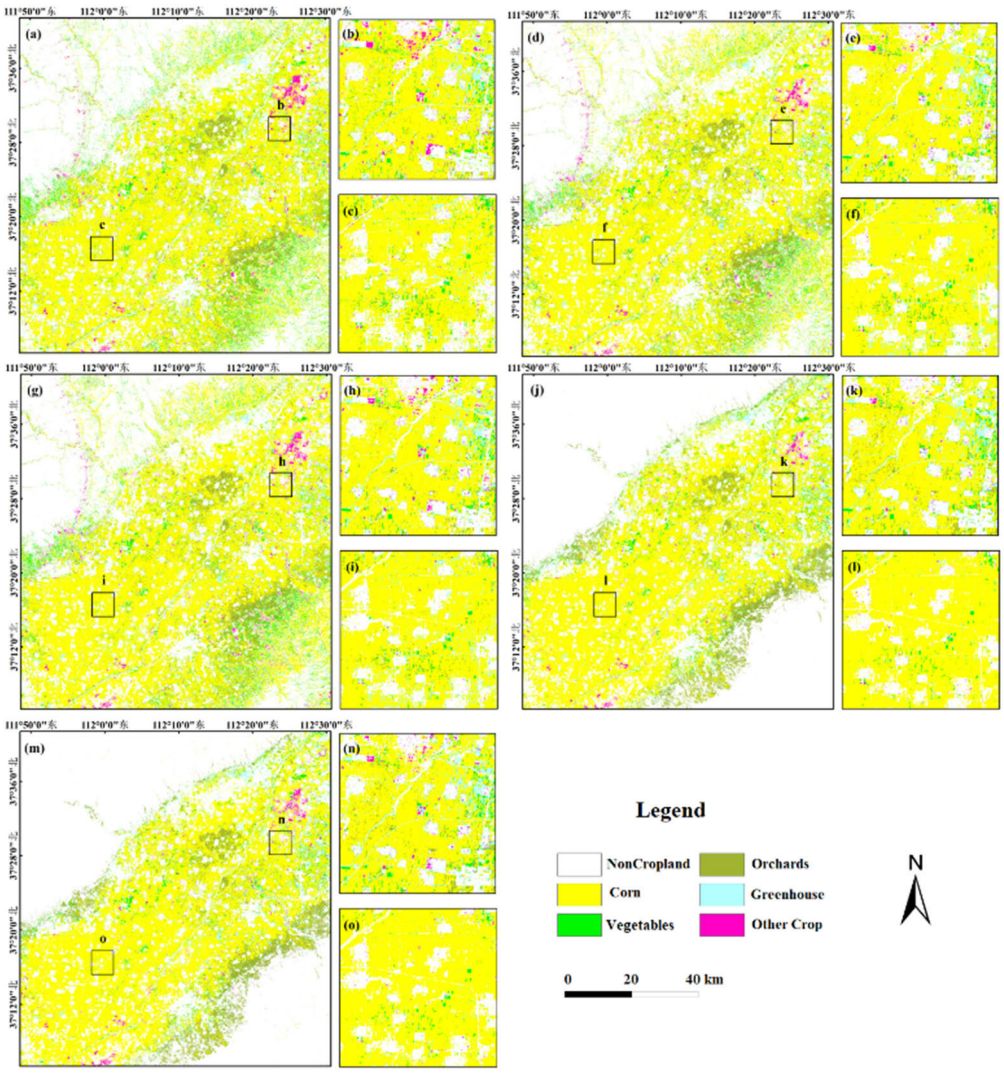



Fig. 8. Crop identification maps of Jinzhong Basin. (a) Result from spectral feature. (b) and (c) Zoom in view of the black boxes of (a). (d) Result from spectral and vegetation index. (e) and (f) Zoom in view of the black boxes of (d). (g) Result from spectral, vegetation index and texture features. (h) and (i) Zoom in view of the black boxes of (g). (j) Result from spectral, vegetation index and topographic features. (k) and (l) Zoom in view of the black boxes of (j). (m) Result from spectral, vegetation index, texture and topographic features. (n) and (o) Zoom in view of the black boxes of (m).

TABLE V

ACCURACY LIST OF CLASSIFICATION RESULTS WITH DIFFERENT FEATURE COMBINATIONS FOR AREA A

| Model | A | B | C | D | E |
|---|---|---|---|---|---|
| Feature combination | Spectral Features | Spectral + Vegetation Index | Spectral+Vegetation Index+Texture Features | Spectral+Vegetation Index+ Topographic Features | Spectral + Vegetation Index+ Texture+ Topographic Features |
| OA /% | 91.9 | 92.5 | 93.6 | 92.1 | 93.0 |
| Kappa /% | 90.0 | 91.0 | 92.0 | 91.0 | 92.0 |
| F1 /% | 88.6 | 88.5 | 90.7 | 87.4 | 90.0 |
| Corn | 92.7 | 94.3 | 94.5 | 94.3 | 94.8 |
| Greenhouse | 87.8 | 91.9 | 92.0 | 92.0 | 92.3 |
| Orchards | 97.8 | 98.1 | 98.1 | 96.9 | 93.1 |
| Vegetables | 85.5 | 88.5 | 89.0 | 89.5 | 91.1 |
| Other Crop | 87.4 | 68.4 | 81.9 | 58.4 | 80.5 |

and 9, where each letter represents one scheme of feature combination, specifically: (a) spectral feature; (b) and (c) zoomed-in view of the black boxes of (a); (d) spectral and vegetation index; (e) and (f) zoomed-in view of the black boxes of (d); (g) spectral, vegetation index, and texture features; (h) and (i) zoomed-in view of the black boxes of

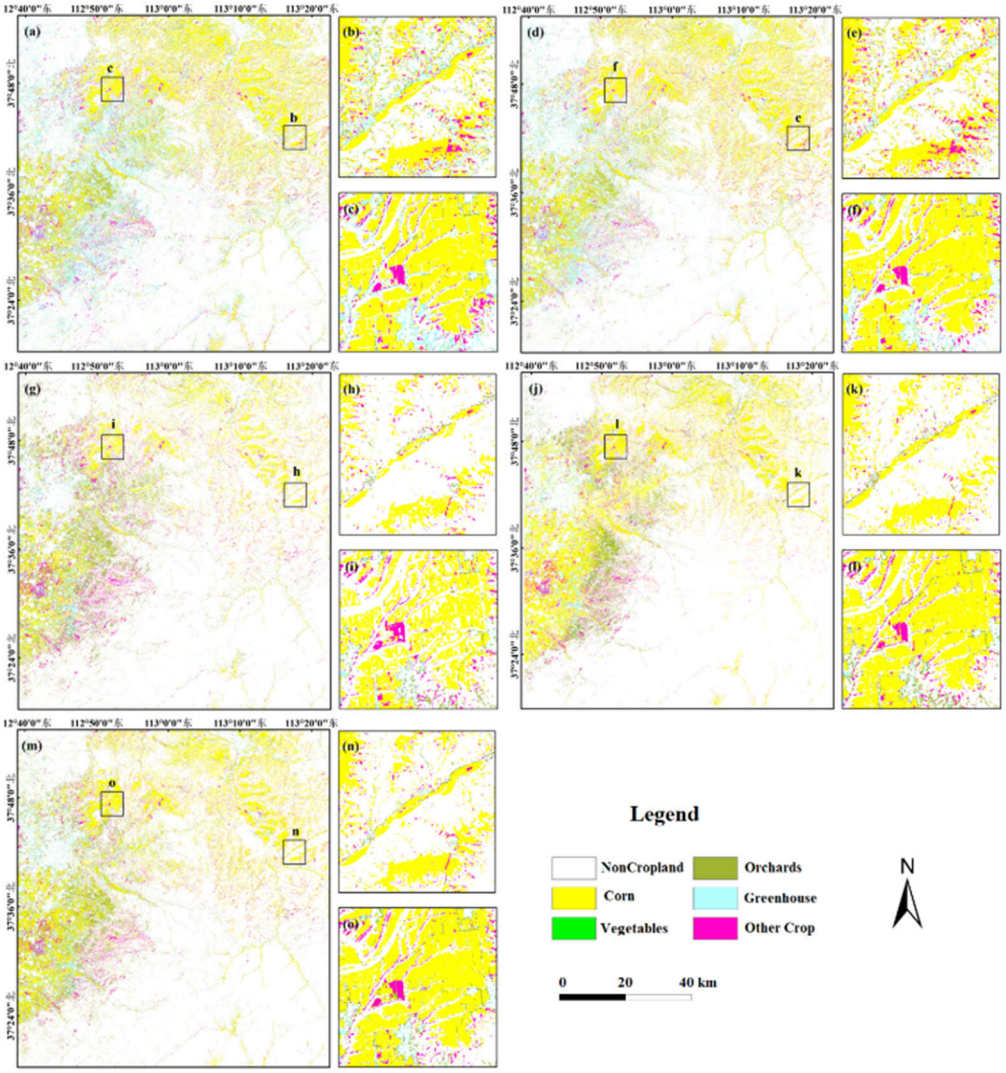



Fig. 9. Crop identification maps of Shouyang Loess Hill. (a) Result from spectral feature; (b) and (c) Zoom in view of the black boxes of (a). (d) Result from spectral and vegetation index. (e) and (f) Zoom in view of the black boxes of (d). (g) Result from spectral, vegetation index and texture features. (h) and (i) Zoom in view of the black boxes of (g). (j) Result from spectral, vegetation index and topographic features. (k) and (l) Zoom in view of the black boxes of (j). (m) Result from spectral, vegetation index, texture and topographic features. (n) and (o) Zoom in view of the black boxes of (m).

(g); (j) spectral, vegetation index, and topographic features; (k) and (l) zoomed-in view view of the black boxes of (j); (m) spectral, vegetation index, texture, and topographic features; (n) and (o) zoomed-in view of the black boxes of (m).

### *B. Accuracy Evaluation of Shouyang Loess Hill*

The accuracy for all proposed feature combinations in Jinzhong Basin is shown in Fig. 10(a), and the OA of all crop types for model C is shown in Fig. 10(b). The accuracy list of classification results with different feature combinations for Jinzhong Basin (area A) is shown in Table V.

In Table V, the different feature combinations show different accuracy. In the case of model A of using the original spectral bands as the feature variable, the OA is 91.9%, the kappa is 90.0%, and the $F1$ is 88.6%. Adding a vegetation index, then the model B was formed, and the classification results achieved a higher accuracy, with OA of 92.5%, kappa of 91.0%, and $F1$ of 88.5%. Continuing to add feature indices resulted in model C with OA of 93.6%, kappa of 92.0%, and $F1$ of 90.7%.

Correspondingly, based on model B, continuing to add topographic features, model D was obtained with OA of 92.1%, kappa of 91.0%, and $F1$ of 87.4%. All three accuracy indicators decreased a bit in comparison with models C or B. Continuing to add topographic features to model C, model E appeared with OA of 93.0%, kappa of 92.0%, and $F1$ of 90.0%. model E achieved slightly higher accuracy than model D but slightly lower than model C.

Finally, model C obtained the highest accuracy with 1.7%, 2.0%, and 2.1% higher than the classification result of spectral feature only, respectively. The classification accuracies for crop types were obtained and listed in Fig. 10(b).

### *C. Accuracy Evaluation of Shouyang Loess Hill*

In Shouyang Loess Hill, the accuracies corresponding to each feature combination in classification are shown in Fig. 11(a), and the accuracy of all crop types within model E is shown in Fig. 11(b). The accuracy of classified results with each feature combination is shown in Table VI.

Fig 11(a) reveals the accuracies corresponding to various feature combinations. Crossing the five models, there are

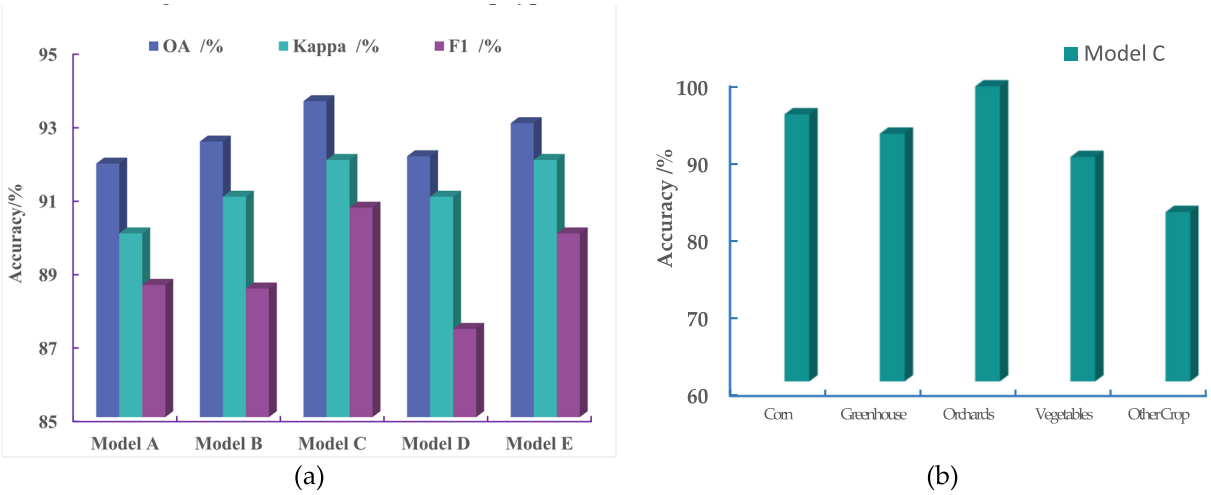



Fig. 10. (a) Accuracies for feature combinations for area A. (b) Accuracy of all crop types for model C.

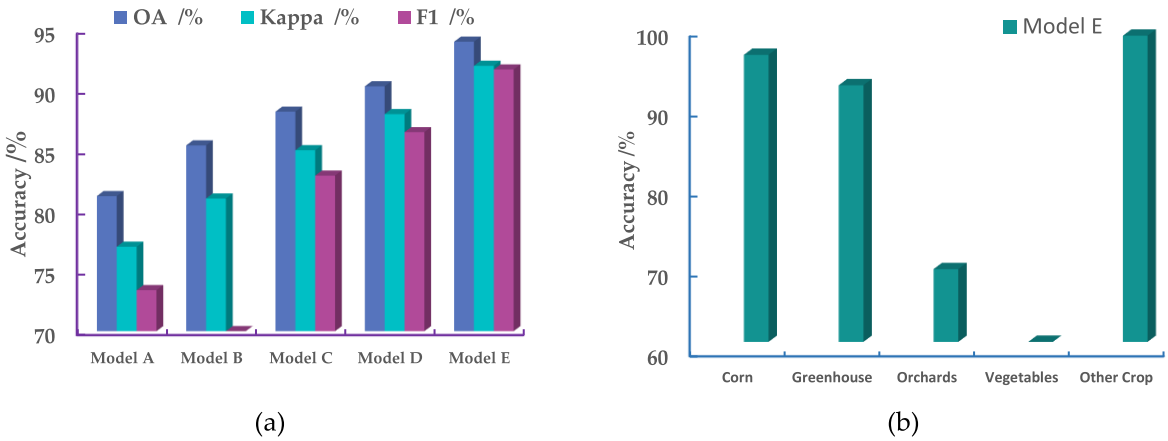



Fig. 11. (a) Accuracies for feature combinations for area B. (b) Accuracy of all crop types for model E.

TABLE VI
ACCURACY LIST OF CLASSIFICATION RESULTS WITH DIFFERENT FEATURE COMBINATIONS FOR AREA B

| Model | A | B | C | D | E |
|---|---|---|---|---|---|
| Feature combination | Spectral Features | Spectral + Vegetation Index | Spectral + Vegetation Index+Texture Features | Spectral + Vegetation Index+ Topographic Features | Spectral + Vegetation Index+ Texture+ Topographic Features |
| OA /% | 81.2 | 85.4 | 88.2 | 90.3 | 94.0 |
| Kappa /% | 77.0 | 81.0 | 85.0 | 88.0 | 92.0 |
| F1 /% | 73.4 | 83. 7 | 82.9 | 86.5 | 91.7 |
| Corn | 87.7 | 87.3 | 89.2 | 86.7 | 95.9 |
| Greenhouse | 64.3 | 87.1 | 91.8 | 90.8 | 92.1 |
| Orchards | 25.9 | 53.3 | 50.3 | 57.2 | 69.1 |
| Other Crop | 89.3 | 94.4 | 93.7 | 94.0 | 98.3 |

existing discernible differences in terms of OA, kappa, and $F1$. In Table VI, in the case of model A, using the original spectral band only as the feature variable, the OA is 81.2%, the kappa of 77.0%, and the $F1$ of 73.4%. When adding the vegetation index, model B was obtained, and the classification results achieved higher accuracy with OA of 85.4%, kappa of 81.0%, and $F1$ of 83.7%, respectively. Continuing to add feature indices resulted in model C with OA of 88.2%, kappa of 85.0%, and $F1$ of 82.9%, respectively. The texture information brought a 7.0% increment of OA, an 8.0% of kappa, and a 9.5% of $F1$ as compared to model A.

Correspondingly, based on model B, continuing to add topographic features, model D was obtained with OA of 90.3%, kappa of 88.0%, and $F1$ of 86.5%. Continuing to add topographic features, model E achieved significantly higher classification accuracy than model D, with OA of 94.0%, kappa of 92.0%, and $F1$ of 91.7%, respectively.

Among these five models, model E was the one with 12.8% increment of OA, 15.0% of kappa, and 18.3% of $F1$, higher than the classification result of spectral features, respectively. In the case of model E, among all crop types, other crops got the highest $F1$ of 98.3%, corn followed with an $F1$ of 95.9%, and greenhouse crops with an $F1$ of 92.1%. Compared to the accuracies of the other three crop types, Orchards is relatively lower with an $F1$ of 69.1% due to less orchard cultivation in this area.

### D. Comparison of Accuracy Evaluation in the Two Regions

Spectral features, vegetation indices, textures, and topographic features are important features for crop identification. However, the topographic features of the Earth's surface vary in different regions, and the impact of these features on the classification accuracy of model results is different. In Table V, among the five models in Jinzhong Basin, there was no significant difference in OA, kappa, and $F1$, and model C got the highest OA, kappa, and $F1$. Adding vegetation index and texture features to the model obviously improved the accuracy, but adding topographic features to the model did not improve the classification accuracy. Therefore, in flat areas, texture features and vegetation indices are the most sensitive and effective indicators that can efficiently improve the classification accuracy, while the topographic features are not sensitive and effective. The inclusion of topographic features has an effect of likely lowering the classification accuracy.

In Shouyang Loess Hill, Table VI illustrates that model E used all feature combinations to extract crop information and achieved the highest OA, kappa, and $F1$. When optimal features are used for crop classification, the key feature information is preserved, and the classification process is improved by avoiding interference of redundant input data. This enhancement has led to significant improvements in various classification types. It can be seen from the results that both texture and topographic factors contribute to the enhancement of the accuracy of crop classification in the loess hill region, while the topographic features play more pronounced roles than texture features. When both were used, there was an amplified effect, and the classification results were further improved. Since the study area is in loess hill areas, topographic feature variables, especially elevation, slope, and other information, can frequently offer significant clues regarding land cover and crop planting distribution in various scenarios. The classification of multispectral images was enriched by using topographic feature variables, and the classification accuracy was greatly improved.

Finally, the inclusion of vegetation index, texture attributes, and topographic elements greatly enhances the classification in complex terrain regions.

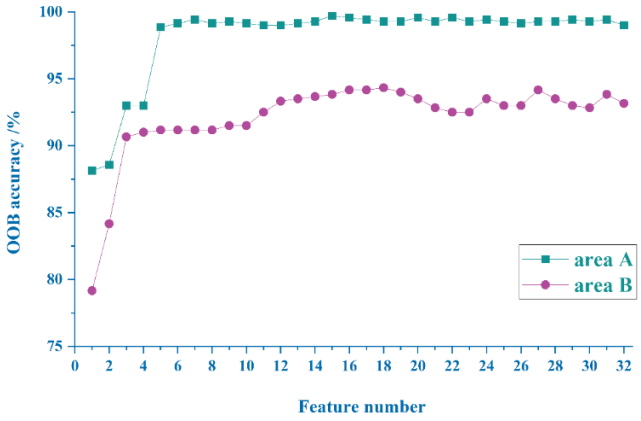



Fig. 12. Relations between OOB accuracy and the number of features.

## V. Discussion

### A. Importance Evaluation of Feature Variables and Selection of Optimal Features

In this study, 70% of the samples were randomly picked up as the training dataset and then used to set up the model based on the RF algorithm. 30% of the remaining samples were utilized as the test dataset to assess the classification accuracy. The OOB accuracy was computed for evaluating the model built with a certain number of features randomly selected from all 32 input features. The importance of features can be computed as the model is built. These features represented spectral features, textural features, topographic features, and vegetation index.

Fig. 12 shows the relations between the OOB accuracy and the number of features in both study areas. The curve for Jinzhong Basin demonstrated that OOB of 99.7% reached its peak when 15 features were selected as input. The curve for Shouyang Loess Hill demonstrated that an OOB of 94.7% achieved when 18 features were selected as input. It does not reach the highest classification accuracy if an insufficient number of features are selected, while excessive and redundant features can cause accuracy to fluctuate instead of improving. Hence, the optimal input features of Jinzhong Basin were chosen by the top 15 features based on their importance rankings, and the optimal input features of Shouyang Loess Hill were chosen by the top 18 features.

Fig. 13 shows the feature importances for two study areas. Based on Fig. 13, the top 18 features for Shouyang Loess Hill and the top 15 features for Jinzhong Basin were selected and used as the input features for the classification with reasonable accuracies and computation complexities, respectively.

As listed in Fig. 13, it is different in terms of the order of the feature importance for study areas A and B. The contributions from each feature for different study areas are also different. Therefore, these features should be used differently and accordingly. In this study, the top 15 crucial feature variables of Jinzhong Basin and the top 18 crucial feature variables of Shouyang Loess Hill were selected as the input to the RF model. Among these optimized features, seven spectral features, four vegetation indices, three texture features, and one topographic feature were used in the Jinzhong Basin area,

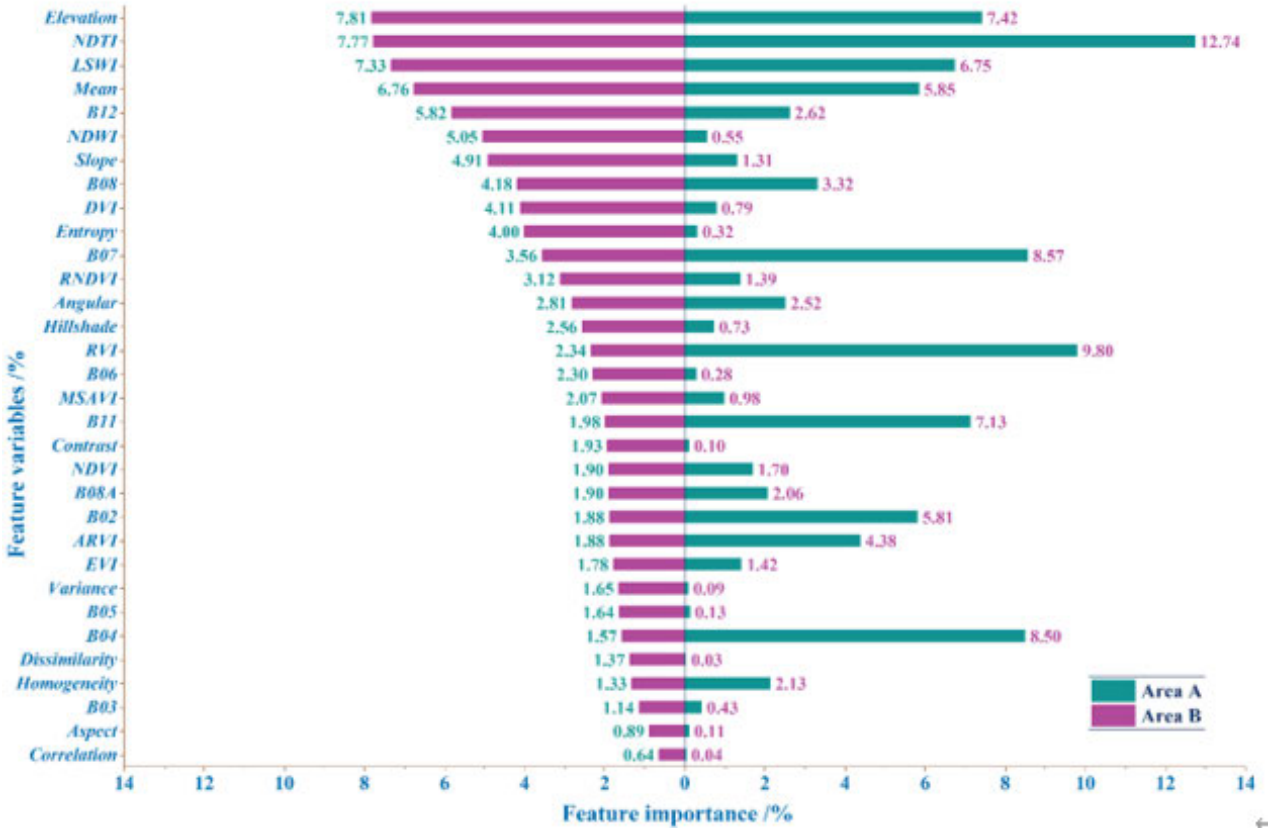



Fig. 13. Feature importance of two types of typical representative regions.

TABLE VII
FEATURE LIST SELECTED FOR THE CLASSIFICATION

| Feature type | Area A | | Area B | |
|---|---|---|---|---|
| | Optimal features | Number | Optimal features | Number |
| Spectral Feature | B02, B04, B07, B08, B08A, B11, B12 | 7 | B06, B07, B08, B11, B12 | 5 |
| Vegetation Index | NDTI, RVI, LSWI, ARVI | 4 | NDWI, NDTI, RNDVI, RVI, DVI, LSWI, MSAVI | 7 |
| Texture Feature | Mean, Angular Second Moment, Homogeneity | 3 | Mean, Entropy, Angular Second Moment | 3 |
| Topographic Feature | Elevation | 1 | Elevation, Slope angle, Hill shade | 3 |
| Total | | 15 | | 18 |

while five spectral features, seven vegetation indices, three texture features, and three topographic features were used in the Shouyang Loess Hill area. The list of selected features may refer to Table VII.

### *B. Comparison of Feature Importance and Optimal Feature in the Two Regions*

According to Fig. 13, in study area A, Jinzhong Basin, of all 32 constructed feature variables, NDTI is the most crucial feature variable, contributing significantly by 12.74%. RVI follows with a contribution score of 9.80%. This underscores the vital role of ratio vegetation indices in extracting crop information in flatland areas. The near infrared band B08A of Sentinel-2 is listed at the 15th place and selected as the last input feature with a contribution score of 2.06%. On the other hand, Dissimilarity has the lowest score of 0.03%, showing that this feature variable has a minor impact on the classification results and can be disregarded.

In study area B, Shouyang Loess Hill, elevation has the highest importance score of 7.81%, indicating that altitude plays an important role in crop identification in a complex terrain area. NDTI, listed in second place, is also the crucial feature variable with a score of 7.77%. The shortwave infrared band B11 of Sentinel-2 is listed at 18th place and selected as the last input feature with an importance score of 1.98%. However, correction has the lowest importance score of 0.64%.

The features, like B04, B02, ARVI, homogeneity, and B08A, played key roles in study area A but were removed for study area B. The features, like NDWI, slope, DVI,

entropy, RNDVI, Hillshade, B06, MSAVI, played key roles in study area B but made less contribution in the study area A. The features, like contrast, NDVI, EVI, variance, B05, dissimilarity, B03, aspect, and correlation played less roles in both study areas.

### C. Analysis of Crop in Complex Terrain Area

The complex terrain area with complex and diverse surface conditions includes valleys, hills, and relatively rugged plateaus, so the classification with remote sensing in complex terrain areas is challenging. According to previous studies, most scholars mainly considered the characteristics of the spectrum itself in crop classification and ignored other features of high-resolution remote sensing images [17], [29]. This study explored a vast array of feature variables. Alongside spectral bands, vegetation indices, texture, and topographic factors were also utilized. Therefore, the significance of the variables ingested by the RF algorithm manifested the priority of feature selection. The results show that additional remote sensing information plays a role when more feature variables are added, and then more accurate crop identification results may be obtained, which aligns with Tao and Hu [30] findings.

The main advantage of this study is filling the gap for crop identification in loess hilly areas with the introduction of new feature variables (elevation, slope, and hillshade), as the topographic features have a certain contribution to remote sensing classification in mountainous areas. Meanwhile, this study also verified the feasibility of the RF algorithm for feature selection and information extraction. Future studies should further explore the potential of a feature selection algorithm that integrates multitime-phase and multifeature approaches, particularly when applied to high-resolution remote sensing images in diverse geographical areas.

## VI. Conclusion

This study focused on crop identification in Shouyang Loess Hill and Jinzhong Basin as case study areas with fragmented field parcels and complex terrains. Sentinel-2 optical images and ALOS DEM data were adopted to extract multiple feature variables, including vegetation indices, texture, and topographic features. An RF model was used to select optimal variables for crop identification and mapping. Five multifeature models were tested, and the results were compared. The findings can be summarized as follows.

1) Spectral information forms the basis for crop classification, and most of the crop information can be captured by spectral features. There is room to further improve the accuracy when more features are used for the classification.
2) The introduction of topographic features and texture features significantly improves crop identification accuracy in complex terrains. In this study, model E shows that OA, kappa, and $F1$ increased by 12.8%, 15.0%, and 18.3% in comparison with only spectral features.
3) In a plain area, additional vegetation index and texture features may improve the classification accuracy a bit. In this study, the optimal features for model C achieved the best performance with OA of 93.6%, kappa of 92.0%, and $F1$ of 90.7%, respectively. However, the additional topographic features did not further help increase the accuracy, but played a negative role when the model ingested both topographic features and texture features.
4) Optimizing feature selection, as shown in this study, is crucial to overcoming dimensionality issues and maximizing the effectiveness of machine learning models for crop mapping.

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